NETWORK PROJECT REPORT

Basic Network Project

VLAN, Inter-VLAN Routing, ACL and Password Security

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Course

Computer Networking

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What is a VLAN?

A VLAN (Virtual LAN) splits one physical network into separate logical networks, one for each department.

1 Group ports

Each switch port is placed in a VLAN, for example VLAN 10 for Accounts.

2 Talk inside

Devices in the same VLAN reach each other directly through the switch.

3 Use the router

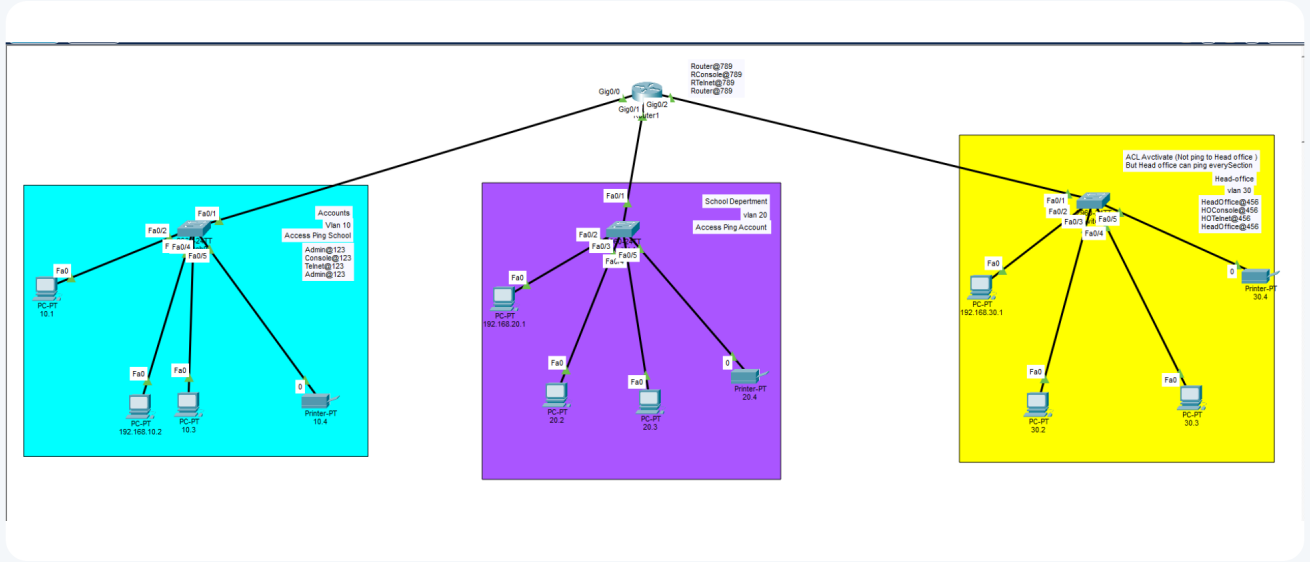
To reach another VLAN, traffic goes to the router gateway, such as 192.168.10.254.

Why use VLANs: better security, less broadcast traffic and easier management.

Network Plan

VLAN	Network	Gateway
10 Accounts	192.168.10.0 /24	192.168.10.2 54
20 School	192.168.20.0 /24	192.168.20.2 54
30 Head-office	192.168.30.0 /24	192.168.30.2 54

One router, three switches. Each VLAN has its own router port. Mask: 255.255.255.0



Router and Switch Configuration

Router1

```
interface gigabitEthernet0/0
ip address 192.168.10.254 255.255.255.0
no shutdown
interface gigabitEthernet0/1
ip address 192.168.20.254 255.255.255.0
no shutdown
interface gigabitEthernet0/2
ip address 192.168.30.254 255.255.255.0
no shutdown
```

Each switch (VLAN 10 shown)

```
vlan 10
name Accounts
interface range fastEthernet0/1 - 5
switchport mode access
switchport access vlan 10

School = vlan 20
Head-office = vlan 30
```

What is an ACL and How It Works

An Access Control List (ACL) is a list of permit and deny rules on a router. It decides which traffic may pass.

1 Packet arrives

A packet enters the router interface where the ACL is applied.

2 Check rules

The router reads the rules from top to bottom.

3 First match wins

The first matching rule decides: permit or deny.

4 No match

Traffic with no match is denied, so we end with permit ip any any.

In this project: ping from Accounts and School to Head-office is denied. Everything else is permitted.

ACL: Block Ping to Head-office

Router1: create the ACL

```
ip access-list extended NO-PING-TO-HEADOFFICE
deny icmp 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255 echo
deny icmp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 echo
permit ip any any
```

Apply it

```
interface gigabitEthernet0/0
ip access-group NO-PING-TO-HEADOFFICE in
interface gigabitEthernet0/1
ip access-group NO-PING-TO-HEADOFFICE in
```

Check it

```
show access-lists
show ip interface gigabitEthernet0/0
```

How Passwords Protect Devices

Passwords stop other people from opening the router and switch settings. Each entry point has its own password.

Enable secret

Protects the privileged mode. It is stored encrypted.

Console password

Protects the console cable port on the device.

VTY password

Protects remote Telnet login to the device.

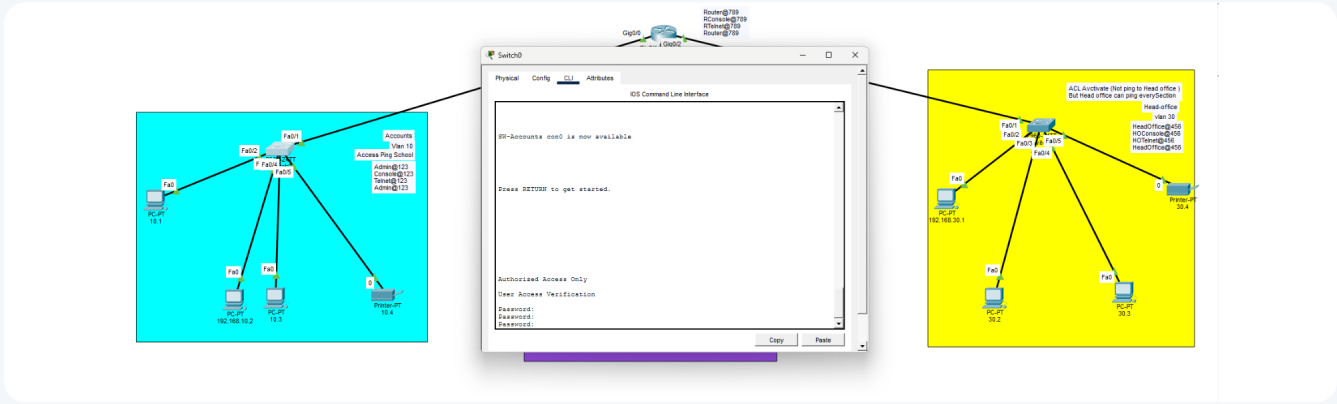
service password-encryption hides the line passwords in the saved configuration.

Password Security

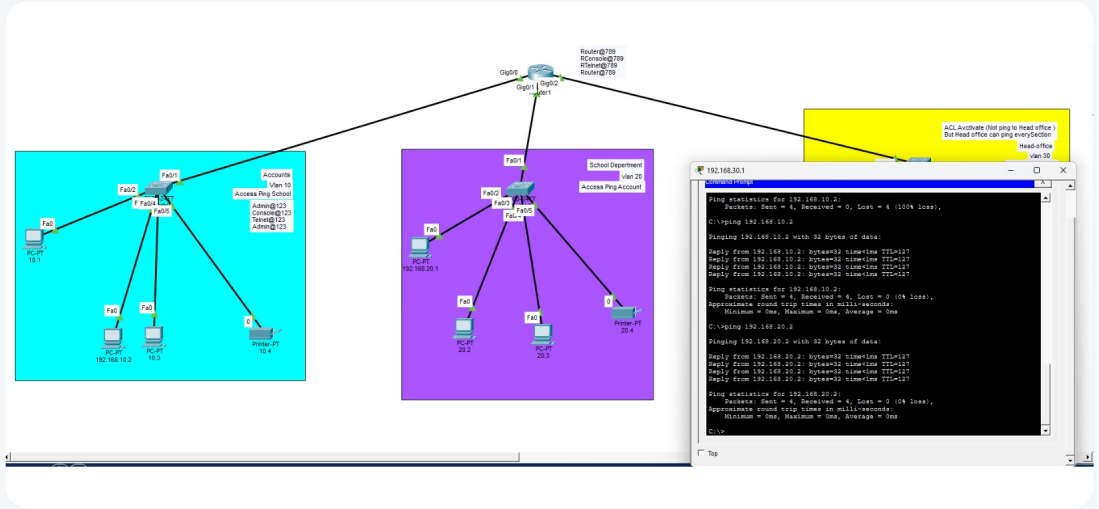
Every device

```
enable secret Admin@123
service password-encryption
line console 0
password Console@123
login
line vty 0 4
password Telnet@123
login
```

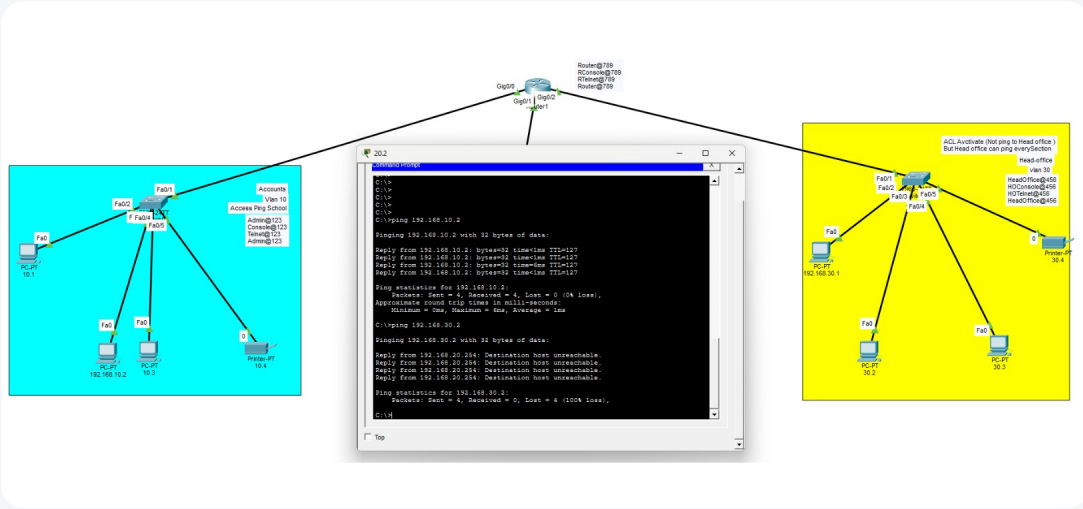
Device	Enable	Console	Telnet
Router1	Router@789	RConsole@789	RTelnet@789
SW-Accounts	Admin@123	Console@123	Telnet@123
SW-School	Admin@123	Console@123	Telnet@123
SW-HeadOffice	HeadOffice@456	HOConsole@456	HOTelnet@456



Test Results



Head-office pings everyone



School cannot ping Head-office

Head-office to all: Success

Accounts and School: Success

To Head-office: Blocked

Troubleshooting and Conclusion

show ip interface brief
router ports must be up / up

ipconfig
check IP, mask and gateway on the PC

show vlan brief
switch ports must be in the right VLAN

ping, show ip arp
ping from the router to find the fault

show access-lists
deny counters rise when blocked

Conclusion

The router connects all three VLANs. The ACL blocks ping to Head-office, and passwords protect every device. All tests passed.